

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Scenario	How the Multimeter Helps
1. Computer not turning on	Measures the power supply voltage to check whether the SMPS is providing the correct output.
2. Smartphone not charging	Checks the charger, USB cable, and battery voltage to identify the faulty component.
3. Laptop battery or CMOS battery issue	Measures the battery voltage to determine whether the battery is healthy or needs replacement.

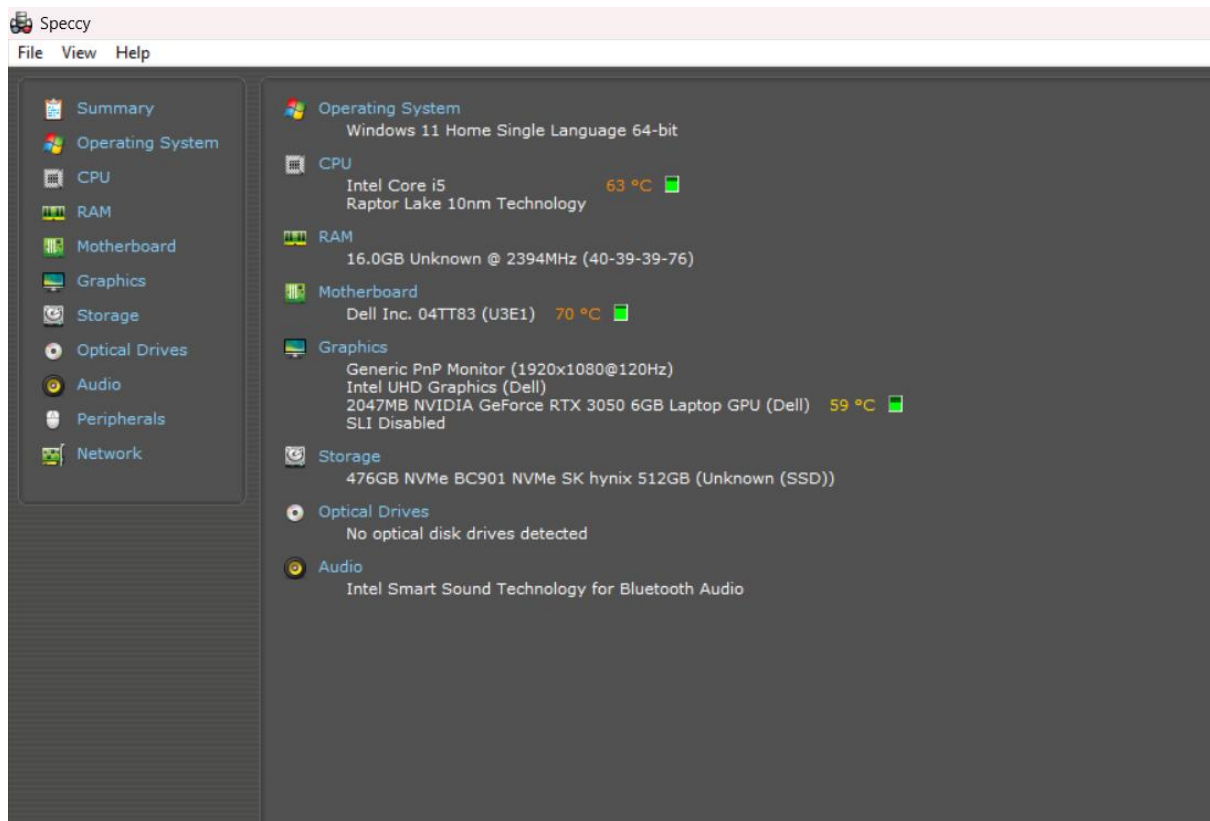
2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.

Ans :

A **POST (Power-On Self-Test) card** displays **two-digit hexadecimal diagnostic codes** while the motherboard starts up. These codes show each step of the boot process and stop at the point where a hardware problem occurs. In the demonstration video, the POST card stopped at a **memory (RAM) error code**, indicating that the RAM was faulty or not properly installed. By reading the displayed code, the technician quickly identified the hardware issue and fixed it by reseating or replacing the RAM.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details. Constraint: Blur or crop out any personal information before submitting.



I installed Speccy, a free system diagnostic tool, and ran it on my laptop. The dashboard displays system information such as the operating system, CPU, RAM, motherboard, graphics, and storage details along with hardware temperatures. No personal information is visible in the screenshot.

4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it Checks	Example Problem Detected
Multimeter	Measures voltage, current, and continuity in electrical circuits.	Faulty laptop charger, dead battery, or SMPS not supplying power.
POST Card	Displays motherboard POST (Power-On Self-Test) error codes during startup.	Faulty RAM, CPU, or motherboard causing the computer to fail to boot.
Software Diagnostics (Speccy/HWMonitor/CrystalDiskInfo)	Checks system hardware information, temperatures, storage health, and performance.	Overheating CPU, failing hard drive/SSD, or low system memory issues.